

Intertwined Practices: Computational Approaches for Handmade Textile Craft

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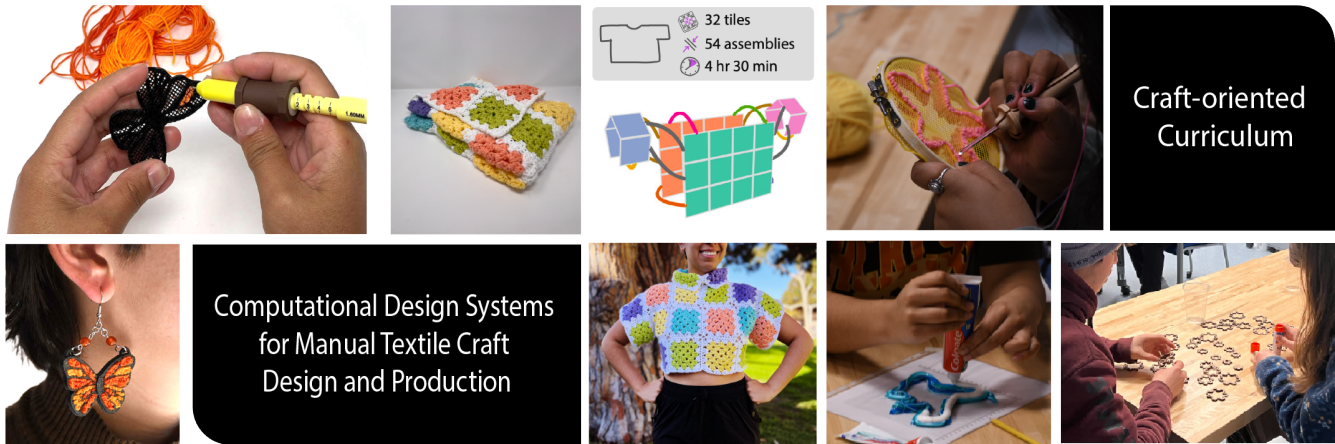


Figure 1: This dissertation proposes works that combine handmade textile craft, digital fabrication, and computational design. It presents two computational design systems, PunchPrint and texTile, and a craft-based curriculum. Future work will develop a programming API and visual design environment for creating custom pixelated charting tools for colorwork in textile crafts.

Abstract

Some forms of handmade textile craft challenge automation due to their structural complexity and the embodied decisions made through movement, repetition, and touch. These practices offer exciting opportunities for HCI to explore how craft-based ways of making and designing can inform new computational tools and fabrication methods. My dissertation explores the intersection of computational design and digital fabrication with handmade textiles. I propose systems in which digital and manual methods operate together (1) to produce new forms of artifacts and computational tools for handmade textile crafts, (2) to preserve and extend the expressive, functional, and social qualities of handwork, and (3) to create more accessible and personally meaningful entry points into computational making for crafters, youth, and creative learners. I developed two digital fabrication systems, which I evaluated through the production of artifacts and workshops with textile craftspeople. I also created a craft-oriented curriculum to engage youth with digital fabrication concepts through manual and hand-machine crafts.

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In the final stage of my research, I plan to investigate how textile practitioners can actively shape their creativity support tools, not as software developers, but as domain experts whose craft practices, values, and material knowledge inform system design.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; • **Applied computing** → **Media arts**.

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1 Introduction

Handmade textiles are created through countless hours of repetitive movements and are often associated with domestic hobbies. Many people practice these crafts as a form of personal enjoyment, care, and creative expression, most commonly by women, and often outside of formal art or design institutions [11]. These forms of making are not aimed at mass production or artistic prestige, but rather at exploring identity, developing material expertise, and creating functional or decorative artifacts. Various textile techniques, including sewing, knitting, and weaving, have played a crucial role in the history of computation and automation. These methods were

instrumental in the development of early mechanical devices, such as the Jacquard loom, and continue to influence modern manufacturing technologies. Because these techniques follow systematic and repeatable steps, they are easier for machines to replicate, effectively translating hand movements into machine instructions. However, not all textile techniques share these replicable behaviors. In crochet, both hands work together to create loops using a single hooked needle. Many of the movements depend on how the individual handles the yarn at that moment. These small choices can vary from stitch to stitch and are influenced by the crocheter's feel, preferences, and experience. Due to this variability, crochet has been difficult to integrate into computational and automated processes, and continues to exist as a purely manual technique. Instead of highlighting the typical advantages of computation and automation, such as efficiency, replicability, and mass production, my research investigates how these technologies can support design and fabrication challenges and limitations present in manual textile craft workflows and materials. By embracing and working within these constraints, my objective is to discover meaningful connections that enhance both handmade textile craft practices and computational methods.

In my thesis work, four projects explore the intersection of handmade textile crafts, digital fabrication, and system design, demonstrating how each can expand the capabilities of the others. First, with my colleague Mert Toka, I developed *PunchPrint*, a technique that integrates fiber-based and filament textiles by combining the 3D printing of a parametric fabric substrate with punch needle embroidery [7]. Second, I developed a youth-oriented CAM-based design curriculum consisting of 1) hands-on activities that introduce machine toolpath concepts, 2) a learner-oriented CAM software interface that scaffolds digital toolpath creation, and 3) project-oriented design activities that integrate CAM-based design and manual craft for ceramics and textile production [8]. Third, I developed *texTile*, a modular fashion workflow that guides designers to make and re-make crochet motifs into versatile tailored garments through digitally fabricated connector pieces and computational design [6]. Lastly, I am currently working on *yarnCode*, where I aim to develop a programming API and visual design environment for creating custom pixelated charting tools for textile crafts. YarnCode contributes to the broader goals of my research by placing the embodied knowledge and material constraints of textile craft at the center of computational toolmaking. This project emphasizes flexible, expressive, and craft-specific tooling that emerges from within the practices themselves.

2 PunchPrint: Integrating Punch Needle Embroidery and 3D Printing

Researchers and designers have used 3D printing to produce structures that mimic the properties of traditional textiles. These techniques enable the fabrication of woven [17], flexible [9], and soft [15] materials and structures through a variety of methods that leverage the versatility of additive 3D printing toolpaths. 3D printing offers programmable control over materials, enabling properties and characteristics that traditional fabrics cannot replicate. Although 3D-printed textiles offer numerous benefits, they lack the softness and natural appearance of traditional fibers. This creates a unique

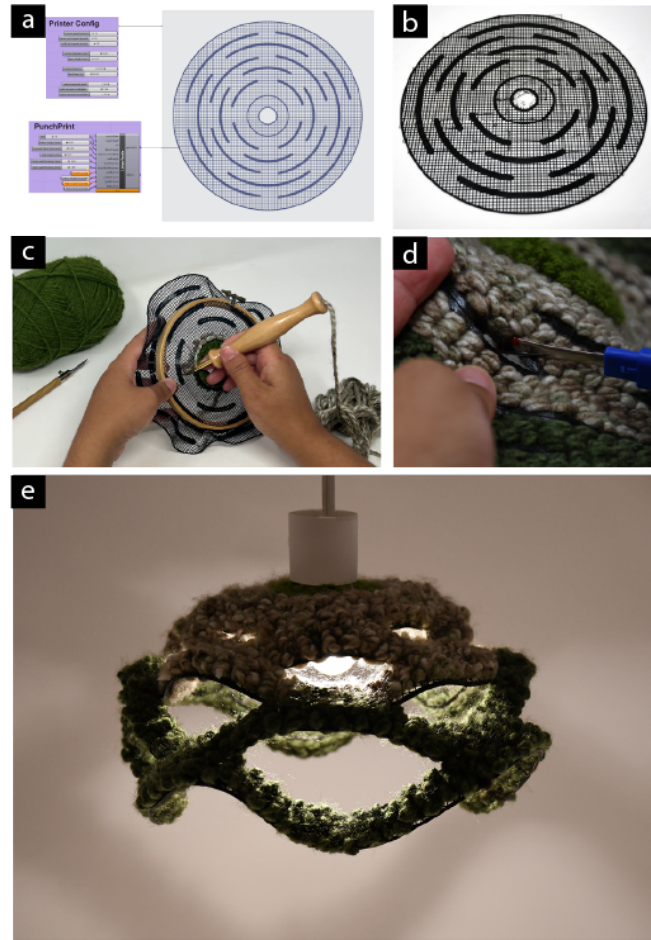


Figure 2: PunchPrint design workflow for punch needle kirigami lampshade: (a) Grasshopper parametric design tool that generates a toolpath for 3D-printing of a textile substrate; (b) fabricated 2D fabric with unopened holes; (c) attaching an embroidery hoop to punch needle the fabric with yarn; (d) cutting the stringing material inside holes with a seam ripper; (e) final artifact assembled with a light bulb.

opportunity for 3D printing and handmade textile crafts, particularly techniques such as punch needle embroidery, to benefit from each other.

Punch needle embroidery is relatively easy to learn and highly expressive. However, it typically requires time-consuming post-processing to secure the edges. To address these challenges, I developed PunchPrint, a technique that integrates filament and fiber-based textiles by combining 3D printing with punch needle embroidery. By substituting traditional woven fabrics with 3D-printed materials, we can avoid the stiffness and fraying often associated with standard foundation fabrics. PunchPrint consists of two main components: (1) a 3D-printed foundation fabric and (2) a parametric design tool integrated into Grasshopper software. The flexibility of TPU fabric allows this grid structure to expand and retract, much as traditional fabric does when a needle is inserted. This means

that the PunchPrint fabric can effectively hold the yarn in place. By combining punch needle techniques with 3D-printed textiles, PunchPrint enhances the potential of both methods. This integration allows 3D-printed textiles to resemble traditional fabrics more closely, while also producing punch needle creations that do not need edge finishing. We evaluated PunchPrint through several examples that exemplify new forms of punch needle embroidery (see Figure 2). Our findings show how, by integrating punch needle methods with 3D-printed textiles, we can expand the applications of both domains.

3 Craft-based Curriculum: Engaging Youth with Digital Fabrication

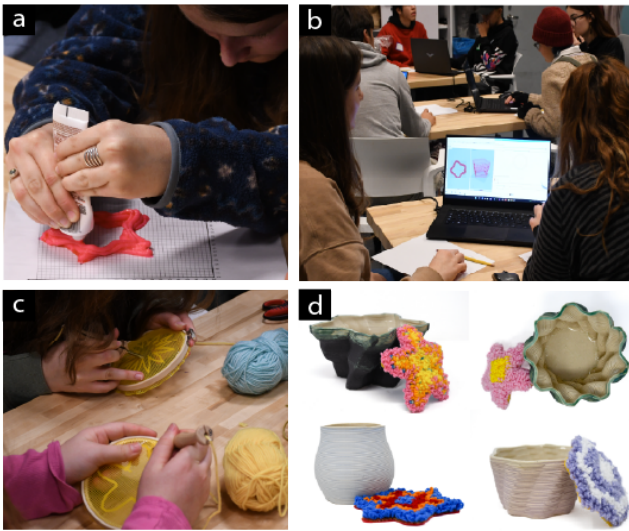


Figure 3: This project presents a craft- and project-based curriculum to engage young people with Computer-Aided Manufacturing (CAM)-based design for additive fabrication. I evaluated the curriculum through a three-day workshop with high school students involving (A) interactive activities, (B) computational design and 3D printing, (C) punch needle embroidery, and (D) the creation of 3D-printed clay and textile artifacts as final projects.

Digital fabrication and computational design offer significant opportunities for introducing powerful concepts, skills, and creative tools to youth [2]. Affordable digital fabrication machines are increasingly accessible in schools, makerspaces, and even homes. Most youth-focused digital fabrication activities rely on Computer-Aided Design (CAD) tools, such as Tinkercad [18], LeoCAD [13], and BlocksCAD [3], to create 2D and 3D representations. In contrast, Computer-Aided Manufacturing (CAM), which involves generating low-level machine instructions (G-Code) to control CNC machines [1], remains largely unexplored as a design method for youth.

In this work, I explore how CAM-based design can support diverse forms of youth expression through craft and digital fabrication. We focus specifically on clay 3D printing and punch needle

embroidery. Clay 3D printing offers a highly visual and tactile experience, making machine behavior and parameter changes easier to grasp. Punch needle embroidery is simple to learn and provides a compelling way to engage with core concepts in both craft and CAM. The curriculum includes three components: (1) hands-on activities that introduce key CAM concepts and the design space of punch needle embroidery; (2) learner-oriented CAM software for designing 3D-printable clay and textile objects; and (3) crafting activities that help students understand and apply design principles relevant to CAM.

To evaluate the curriculum, we led a three-day workshop with 12 high school students. The students explored additive fabrication and layer manipulation using tangible materials. For example, they used laser-cut cardboard and glue to build layered forms and experimented with frosting tubes, toothpaste, and spray cheese to simulate nozzle-based extrusion. These playful exercises helped them explore key concepts such as toolpath design, nozzle size, material behavior, and G-code generation. Participants reported that these activities improved their understanding of how 3D printers work and clarified the relationship between material, toolpath, and machine behavior. Our findings suggest that digital fabrication can be taught without expensive tools or complex software. Familiar craft materials and accessible fabrication techniques make learning more inclusive and engaging. This work demonstrates that young people can effectively develop foundational digital fabrication skills through hands-on, material-centered, and craft-informed experiences.

4 texTile: Integrating Modular Fashion into Handmade Crafts

Modular fashion is a design approach that emphasizes flexibility, customization, and sustainability by creating clothing items that can be easily re-assembled, resized, and re-purposed [4, 14]. Although modular fashion has significant potential for creative and sustainable garment design, its practical implementation remains challenging. Commercial “modular garments” typically feature detachable components, such as sleeves or panels [5, 16]. These designs allow consumers to mix and match pieces to adapt to changing needs. However, there are relatively few commercially available forms of modular fashion, and those that exist are often limited to simple variations of the same object, for example, converting pants into shorts. Artists and fashion designers have experimented with constructing garments using interlinked modules that can be reconfigured into various styles [10, 12, 19]. This building-block approach allows artists to explore a wide range of design possibilities, but current methods often rely on specialized machines and bespoke manual techniques for assembly.

While modular fashion has gained traction in contemporary design, similar principles have long been integral to textile craftsmanship. Traditional crafts like quilting, patchwork, and crochet granny squares constitute modular design methods by combining smaller elements into larger, intricate patterns. A granny square is a textile motif created with crochet, made individually and then joined together to form larger textiles like blankets or garments. Although granny squares embody a modular design approach, traditional joining techniques are arguably permanent since cutting a

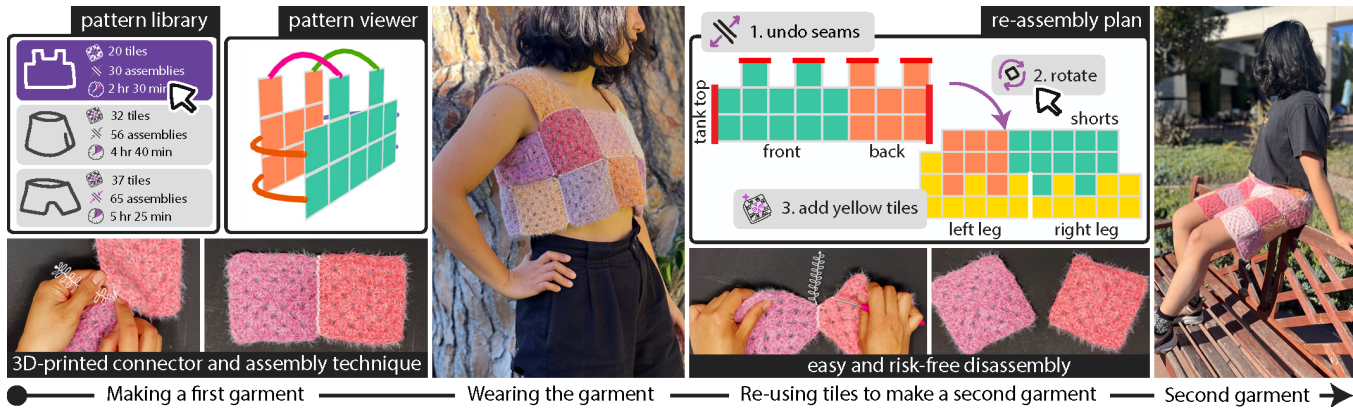


Figure 4: *texTile* is a workflow for modular tile-based garment construction within the crochet granny squares technique. The *texTile* workflow supports making and re-making garments by re-using crochet tiles. This workflow leverages tools from computational fabrication and computational design. It has two components: a non-permanent assembly technique using 3D-printed connectors, and a software tool with a pattern library, pattern viewer, and re-assembly plan.

yarn joint without accidentally damaging the motifs can be difficult. Despite partaking from a modular design practice, crochet granny square craft lacks well-established workflows that enable motif re-use. The challenges with traditional granny square methods stem from three main issues. First, these techniques, using a tapestry needle or crochet stitch, permanently attach motifs, making disassembly difficult and risky. Second, creating garments requires a specific pattern, but finding or adapting versatile patterns for modular use can be challenging. Lastly, modular fashion relies on the ability to deconstruct and reassemble garments, but crocheters may hesitate to undo their work out of fear of damaging the motifs. These factors limit the modular potential of granny squares and can undermine a crocheter’s confidence in reusing their creations.

To address these challenges, I developed *texTile*: a modular fashion workflow that guides designers to make and re-make crochet motifs into versatile tailored garments through digitally fabricated connector pieces and computational design. The 3D-printed connectors are flexible strips sized to fit the edges of granny squares. They can be assembled using a crochet hook with a technique similar to a crochet zigzag stitch. The objective was to simplify the process for crafters to reassemble tiles into a new garment without having to disassemble everything completely. To accomplish this, the system calculates an efficient method to reuse as many seams and tile placements as possible, minimizing the number of tiles that need to be detached and reattached.

Once the second design is chosen, the system provides a step-by-step reassembly plan. Utilizing the color-coded tiles from the first pattern helps the user identify which tiles to move and where to place them in the new design, making the process of transforming one garment into another more efficient, but also clearer and less intimidating. The system was evaluated through a study involving four experienced crochet and garment makers. In the user study, we observed that participants quickly learned the connector technique. They found the *texTile* library to be helpful, appreciating that they could use it creatively rather than feeling obligated to follow the patterns exactly. Additionally, seeing reassembly options

helped participants envision new garments and made them more comfortable taking apart the first one.

5 Future Work

In the course of developing this fourth project, I aim to explore how computational abstractions can support a wide range of making practices while remaining accessible and meaningful for crafters. I currently refer to this project as *yarnCode*, a programming API and visual design environment to create custom pixelated charting tools for textile crafts. Pixelated charts are composed of small squares, similar to pixels, that guide the making of pictorial textile patterns. These charts typically feature numbered axes, symbols, and structures that illustrate the logic of the specific textile technique for which they were created. Crafters commonly use digital tools to design these charts. Some patterns are drawn manually, square by square, which offers precise control but makes it difficult to iterate. Others rasterize images to generate charts automatically, though this often requires adjusting multiple parameters to produce a usable result. Although these charts may appear simple, they contain essential information that shapes the physical making process.

Pixelated charts serve as a shared visual language in many handmade textile crafts, including cross-stitching, crochet, knitting, and bead weaving. Each square in the chart corresponds to a specific action, such as a stitch, a color change, a bead placement, or a weaving maneuver. Despite their importance, most digital tools treat pixelated charts as static patterns instead of dynamic, customizable systems. This raises the question: what if crafters could define how charts behave, adjusting logic, rules, or constraints depending on their technique and materials? *YarnCode* is a system of stackable functions that turn visual designs into fabricable charts, customized for each craft by the crafter. These functions enable crafters to evaluate and adjust their pixelated charts, ensuring they align with the physical actions of their craft and any constraints set by the maker. For example, in knitting, a crafter can set a maximum yarn float length, and the system will scan the design, highlight where

the rule is broken, and optionally adjust the chart to meet that limit. This project is guided by the following research questions:

- RQ1: How can a modular API be designed to support diverse textile pixelated charting techniques through craft-specific logic?
- RQ2: What computational abstractions enable meaningful manipulation of pixelated charts across different crafts?
- RQ3: How do different crafters conceptualize and interact with pixelated patterns, and how can these preferences shape our API and UI design?

YarnCode contributes to the broader goals of my research by placing the embodied knowledge and material constraints of textile craft at the center of computational toolmaking. This project focuses on creating expressive, craft-specific tools grounded in the knowledge and expertise of crafters.

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